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Problem Chosen

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2006 Mathematical Contest in Modeling (MCM) Summary Sheet

(Attach a copy of this page to each copy of your solution paper.)

Type a summary of your results on this page. Do not include
the name of your school, advisor, or team members on this page.

Summary

In this paper, we develop a series of models to predict the trend of AIDS infections and give suggestions on how to control it and manage financial resource effectively.

According to the threat of AIDS, we choose a country in each continent to investigate.

Firstly, we propose a flexible model to predict the total number of HIV/AIDS infections in the next 45 years for six countries. The model is adaptive for three cases. By analyzing a large amount of data, we find that infections in different gender and age play different roles in the spread of HIV. Then, we divide people into different groups by gender and age every 5 years. Based on the analysis of mechanism of transmission of HIV, we propose a recursion equations model, which considers the transition of different groups, to predict the number of HIV infections every 5 years.

In the absence of any additional interventions, we utilize the data from 1981 to 1992 to calculate the parameters namely death rate, infection rate. The parameters are applied to the recursion equations model to estimate the number of HIV infections in the following 45 years. Based on the model, we find that the percentage of AIDS in population grows earlier and decrease later gradually.

In order to estimate the level of financial resources, we investigate the affection of prevention interventions and care interventions to the growth rate of AIDS infection. By a continuous differential equation model, we utilize a qualitative analysis for solution to differential equations and then conclude that the growth rate of AIDS infection can be reduced by both prevention and care interventions. But the number of AIDS patients can only be reduced by prevention interventions. Based on the previous conclusions, we determine the financial resources for different countries, according to the GNP of these countries and the cost to interventions

Taking ARV-resistant strains into account, we give the method to deal with the problem.

At last, we write a white paper providing our recommends to UN about HIV/AIDS.

Fighting against AIDS

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Introduction

Since June 1981, the first cases of what is now known as AIDS were reported in the USA, people have struggled with HIV/AIDS for 25 years. Up to the present, both the number of infections and number of deaths due to the disease continue to rise, more than 25 million people have died of AIDS up to 2005.

In order to control the spread of HIV/AIDS effectively and try to reduce it to the minimum, we should monitor and estimate the spread of the world epidemic and to assess the need for services and resources.

Therefore, the first thing to be done is to approximate the number of HIV/AIDS infections.

For each continent (Africa, Asia, Europe, North America, Australia, and South America), we choose a country to represent its continent to analyze by calculate its evaluating indicator, which is determined by the number of health person, infections and the GNP of the country.

Then we should develop a flexible model, which can adapt to various situations. For different cases, what we need to do is just to adjust certain parameters.

Regardless of any additional interventions, as the period that dated from 1981 to 1992, is the initial time that AIDS came into public view, no effective measures could be taken to prevent it from spreading. So we choose the data of these years to calculate the parameters we need.

There are two effective approaches that can control the spread of HIV/AIDS, use antiretroviral (ARV) drug therapy to control the source of infection and use preventative HIV/AIDS vaccine to help prevent HIV infection. Utilizing available financial resources, how to adjust the application of these two approaches in each country to achieve the best effect is an important affair. There are three optional scenarios:

1. ARV drug therapy
2. A preventative HIV/AIDS vaccine
3. Both ARV provision and a preventative HIV/AIDS vaccine

For each the six countries, there are two aspects of spending should be considered: infrastructural investment and the cost of therapy and prevention. After analyzing the situation of each country, we select an optimal scenario to implement for each country.

Now as the status of each country has been determined, the parameters can be determined too. So the rate of change in the number of HIV/AIDS infection of each country can be approximate too.

If fact, the effect of ARV-resistant disease strains could not be neglected, we should adjust parameters again so as to make our model more realistic.

Finally, according to the results of upper models and other data we give some recommendation on controlling the spread of AIDS.

Background

Before beginning the fight against HIV/AIDS, let's take in some background about HIV and AIDS.

What is AIDS

AIDS is the short name for Acquired Immune Deficiency Syndrome, a disease that weakens a person's immune system. People with AIDS are much more likely than other people to catch other diseases besides AIDS. These other diseases are called opportunistic infections (OIs for short), because the weakening of a person's immune system gives these diseases the opportunity to infect the person. AIDS is caused by Human Immunodeficiency Virus, which is called HIV for short.

What is HIV

HIV is the short name for Human Immunodeficiency Virus, the virus that causes AIDS. A person who is infected with HIV is said to be "HIV positive." HIV doesn't go away on itself, and there is currently no cure for HIV.

How can AIDS be transmitted?

There are three kinds of propagation path of AIDS in all: Sexual contact, blood contact and mother to child transmission.

You can get HIV from: Vaginal sex, Oral sex, Anal sex, Injecting drugs, Blood transfusions, Blood products, Mother to child transmission, Infection in the health-care setting and Tattoos / piercings.

Many people cannot distinguish the differences between being HIV positive and having AIDS. As most literature makes a distinction between people who are infected with HIV but are still healthy, and those who have the virus and are sick with AIDS. This may lead readers to think that these are two different diseases, but this is not true. They are simply two different stages of the same disease. Now let's get to know the differences between these two aspects.

The difference and relation between being HIV positive and having AIDS

First, being HIV positive and having AIDS are two different conceptions. A person is said to be HIV positive if that person is infected with the virus HIV. The person may not be sick at all. A person is said to have AIDS if the person is infected with HIV and has certain signs and symptoms of the disease. In short, HIV is the virus, and AIDS is the disease that it causes.

Any person who has HIV, whether the person is sick with AIDS or not, should be under the care of a qualified physician. Now, more than ever, there are treatments that can delay the onset of the disease. The earlier one seeks treatment, the better one's chances are likely to be. An infected person may seem very healthy at first. But without treatment, most infected people start to get sick a few years after first

becoming infected.

Without drug treatment, HIV usually progresses to AIDS in an average of ten years. There have been a few rare cases in which people have been infected for over a decade without treatment and without showing any signs of AIDS.

Any person who has HIV, whether the person is sick with AIDS or not, can pass the virus on to other people. This means you can get AIDS from a person who looks very healthy, if the person is infected with HIV and if you have sexual or blood contact with the person. [Lisa Murphy 1998]

Control the spread of HIV/AIDS epidemic

We are aimed at constraining infection rate and limiting negative effects of HIV/AIDS. To meet the requirement, we developed a flexible and universal model, which can be applied to various cases by changing or increasing certain parameters.

Therefore, we start studying the problem about HIV/AIDS with the characteristics of the spread of the epidemic (AIDS). To solve problems about HIV/AIDS in mathematic way, we set up some necessary parameters according to different conditions. For different cases, different value of parameters should be considered.

Preparative

- **Divided the people in one country by their gender and age (every 5 years) into different groups.**

(1) The proportion of male sufferer is higher than female sufferer at large in most countries.

On worldwide, by December 2005 women accounted for 46% of all adults living with HIV worldwide.

In addition, the causes of male and female HIV carriers are different.

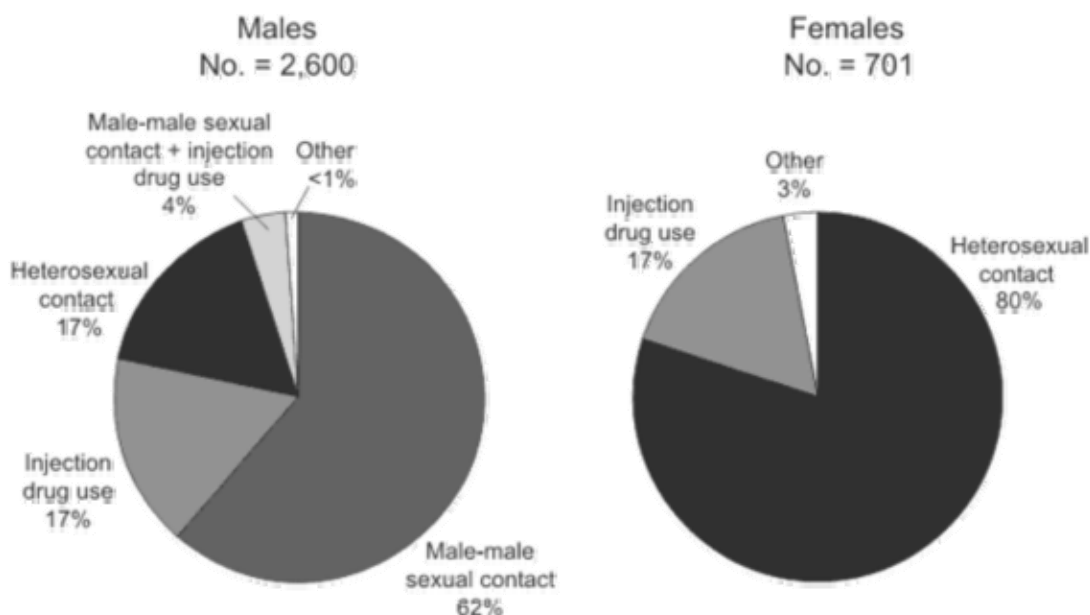


Figure 1. Estimated cases of HIV/AIDS diagnosed in 2002 [CDC]

From the figure it appears that the causes of male and female infections vary greatly.

(2) The prevalence rates differ from each other in different time of life.

Young people (15-24 years old) account for half of all new HIV infections worldwide - more than 6,000 become infected with HIV every day. People living with HIV/AIDS aged 25-59 are comparatively rare. Minor infections (aged 0-14) are much less. There are hardly any senile infections aged upper 60. [AVERT 2005]

Ground on the two phenomenon upwards, we consider dividing people into some groups by gender and age (every five years).

We name males aged 0-4 “male group1”, 5-9 “male group 2” etc.

We name females aged 0-4 “female group 1”, 5-9 “female group 2” etc.

Definition

Effectual contact rate: The number of person that HIV/AIDS infections contiguity with effectually every five years. That is to say, when a HIV/AIDS infections contiguity with a healthy person effectually, the healthy person would be infected.

Assumptions

1. All data are actual, practical and trusty.

There is often a delay between the time of diagnosis of HIV or AIDS, or the time of death, and the time at which the event is reported. Moreover, this delay may differ among different categories of people. In addition, reliable data on HIV prevalence rates by county are generally difficult to obtain, most of the data are estimated. So, to simplify our model, we assume that when certain organization estimates the number of diagnoses, deaths and people living with HIV or AIDS have considered the effect of those factors and redistributes cases into exposure categories if none was initially reported. All data are estimated by adjusting for reporting delays, taking into account the differences between categories. No adjustment is made for incomplete reporting.

2. People in one group are just under similar conditions to each other.

To simplify our model, when analyzing people's infection percentage, resistivity and other factors, we assume that these factors are the same to people in one group.

3. We just need to consider AIDS diagnoses infections from 0 to 59.

There's no need to take senile HIV infections into account because there are hardly any senile infections aged upper 60, and additionally, latent period of AIDS is 10 years on average, those senile infections may die after 10 years (the time that AIDS outbreak) even they are healthy.

4. The number of the population is static in five years.

There isn't much change in the number of population on worldwide in 5 years. We assume it's static to simplify the model.

5. The number of newborn infants who contract the AIDS of each gender is the same.

According to biologic knowledge, we know that the proportion of man to woman in newborn is 1:1, and the fact won't be changed by the fact that their mother is a HIV carrier.

7. Ignore the number of immigration and emigration.

The number of immigration and emigration is very little relative to the total number of a country.

Symbols used in this paper

S	The evaluating indicator to evaluate the spread of a country's epidemic
$A(n)$	Number of people living with HIV/AIDS
$A_N(n)$	New AIDS diagnoses infections
$P(n)$	Population
$G(n)$	Per-capita gross national product (GNP)
n	Year
$b(n)$	Birth rate
$\eta(n)$	Death rate of infections
γ	The probability of a person be infected by HIV/AIDS
$F(n)$	The number of women with the capability of procreating
$\theta(n)$	Fertility rates
$\rho(n)$	
λ	Effectual contact rate
$i(n)$	The proportion of infections
$h(n)$	The proportion of healthy person

Select a country for each continent

Although continents develop to totally different degree, for they each in the same continent, the economic and health situations in most of the countries are similar. Regarding of this, we can chose one critical country from each continent to analysis the spreading speed of AIDS in there.

To select a critical country for each continent, an appropriate evaluating indicator must be established at first.

There are three main factors affect the spreading speed of AIDS that we take into account:

- 1) The number of people living with HIV/AIDS which we mark as $A(t)$.
- 2) Population
- 3) GNP (per-capita gross national product)

With assumptions that:

- 1) As AIDS is a kind of infectious diseases, the initial number of AIDS carriers has a great impact on the number of AIDS carriers in the future time, and this impact is positive.
- 2) The number of potential carriers is closely related to the number of non-carriers, and this impact is also positive.
- 3) GNP per capita data give an accurate indication of the dollar value of a nation's economy, it can reflect how wealthy a country is to a certain degree. A country with a higher GNP will

make its public sanitary fixture more healthy, give its subjects better education, and will take better precautions against AIDS, so the number of AIDS carriers in a certain number of citizens is smaller.

Based on these assumptions, we establish an evaluating indicator S to evaluate the spread of HIV/AIDS of a certain country. Express by formula is show as follows:

$$S = \frac{A(t) \times (P(t) - A(t))}{G(t)}$$

This formula shows that along with the increase of the number of people living with HIV/AIDS and the number of healthy people, the condition is more serious. But the condition will alter for the better along with the increase of GNP.

We can interpret the problem is to seek a country whose S is the largest in the continent. To reach our goal, the first step is data processing. Utilize the data from WHO, we figure out S for each country in 1999. The result is given in Appendix 1. Then we select some countries of each continent whose S are larger than any others. In these countries, we selected the most critical one according to other factors additionally.

Finally, we select these countries: America for North America, China for Asia, Australia for Australia, Nigeria for Africa, UK for Europe and Mexico for South America.

Recursion model to predict the number of HIV/AIDS infections

First of all, analyzing the problem, we should predict the number of HIV/AIDS infections in future.

- 1) To solve this problem, we break it down into some certain subproblems. That is, to appropriate the expected number of HIV/AIDS infections of each group which we have partitioned.
- 2) The number of new AIDS diagnoses infections is related to the total number of infections and healthy person.
 - a) The number of people who are contact with infections increase along with the increasing of the number of infections meanwhile the probability of a person infected increase along with the increasing of healthy person. as a result of lacking of vigilance.
 - b) On the other hand, the number of new infections would be reduced with the decrease of infections because the chance of contact is reduced.
 - c) It would be reduced with the decrease of healthy person.

There's an upper limit of the number of infections. Most of the HIV carriers are infected because of their profligate behaviors, the number of susceptible is limited. Additionally, people's vigilance would increase along with the increasing of infections.

This relation can be described by following formula:

$$A_N(n) = \rho(n) \cdot A(n) \cdot (P(n) - A(n))$$

Footnote: $\rho(n)$ is a proportional action coefficient. We would recommend how to determine it later.

- 3) When estimating the number of people infect of each group (aged 5-59), we just should consider two aspects: the number of new AIDS diagnoses infections and the number of people who are in younger age group five years ago that still living and grow up.
- 4) Estimating the number of infantile infections aged 0-4 is different from the approach of

estimate the number of infections aged 5-59. These infant are infected mainly though Mother to child transmission.

5) The fertile activity would be reduced by 20% if a female were infected. [AVERT]

6) 25% of the newborn infant would be infected though mother to child transmission. [AVERT]

The number of HIV/AIDS infections of each group (expect infections aged 0-4) can be calculated by following formula:

$$A_{mi}(n) = A_{Nmi}(n) + A_{mi}(n-5) \cdot (1 - \eta(n-5))$$

$$A_{Nmi}(n) = \rho_{mi}(n) \cdot A_{mi}(n) \cdot (P_{mi}(n) - A_{mi}(n))$$

$$A_{fi}(n) = A_{Nfi}(n) + A_{fi}(n-5) \cdot (1 - \eta(n-5))$$

$$A_{Nfi}(n) = \rho_{fi}(n) \cdot A_{fi}(n) \cdot (P_{fi}(n) - A_{fi}(n))$$

The number of HIV/AIDS infections aged 0-4 can be calculated by following recursion:

$$A_{m1}(n) = 0.5 \cdot (1 - 20\%) \cdot 25\% \cdot \sum_{i=4}^{10} (F_i(n-5) \cdot \theta_i(n-5))$$

$$A_{f1}(n) = 0.5 \cdot (1 - 20\%) \cdot 0.25\% \cdot \sum_{i=4}^{10} (F_i(n-5) \cdot \theta_i(n-5))$$

Parameters explanation:

$A_{mi}(n)$: The number of male group i in year n .

$A_{fi}(n)$: The number of female group i in year n .

$A_{Nmi}(n)$: The new number of AIDS diagnoses infections of male group i in year n .

$A_{Nfi}(n)$: The new number of AIDS diagnoses infections of female group i in year n .

$\eta(n)$: the average death rate of infections.

$F_i(n)$: The total number of women with the capability of procreating of female group i in year n

$\theta_i(n)$: The fertility rates of female group i in year n .

We alter the value of certain parameters according to different cases, then the number of HIV/AIDS infections from 2006 to 2500 can be approximate calculate, and so is the rate of change in the number of HIV/AIDS infections.

Estimate the number of infections in the absence of any additional interventions

Before applying our model to predict the HIV/AIDS infection for the selected countries from 2006 to 2050, we must determine some parameters such as death rate, infection rate and so on.

Regardless of any additional interventions, as the period that dated from 1981 to 1992, is the initial time that AIDS came into public view, no effective measures could be taken to prevent it from spreading. So we choose the data of these years to calculate the parameters we need.

Some parameters such as death rate, birth rate can achieve easily from data directly or by simply calculation, the others are hard to get which should be calculate by analogy.

Take America for example, we determine the value of parameters as follows:

1. Death Rate

Firstly, we estimate the death rate every five years because our model regards the five-year as a whole. **Table 1** shows some necessary data:

Table 1
Cases diagnosed & Deaths occurring every year

Year	1981	1982	1983	1984	1985	1986	1987	1988	1989
Diagnosed	339	1201	3153	6368	12044	19404	29105	36126	43499
Deaths	130	466	1511	3526	6996	12183	16488	21244	28054
Year	1990	1991	1992	1993	1994	1995	1996	1997	1998
Diagnosed	49546	60573	79657	79879	73086	69984	69124	49379	43225
Deaths	31836	37106	41849	45733	50657	51414	38074	21846	19005

We also have the total cases at the beginning of 1981, that is 70. It is clear that we get the following equation:

$$S(n+1) = S(n) + N(n) - D(n)$$

Where $S(n)$ is the total cases at the beginning of year n , $N(n)$ is the diagnosed cases during that year, and $D(n)$ is the deaths in year n .

We define the death rate $R(n)$ by average:

$$R(n) = \frac{D(n)}{S(n) + N(n)}$$

So, in our model we need five-year death rate. Similarly, we also have the five-year death rate R_i :

$$R_i = \frac{\sum_{k=1}^5 D(k)}{S(n-4) + \sum_{k=1}^5 N(k)}$$

So we apply the MATLAB program to determine R_i . To simplify the problem, we define the average death rate R that is the mean of all R_i for all i . Lastly, the R is 0.40.

2. Infection Rate

Based on the assumption 2, we divide the whole people into 13 groups, i.e. from 0 to 59 (five years group), and 60 plus. However, the data we searched is not desired. For example, the data about USA in year 1986:

Table 2

Cases diagnosed by age group in year 1986

Age	Under 5	5-12	13-19	20-29	30-39	40-49	Over49
Cases	361	49	127	6077	13574	5986	2829

In order to fulfill our basic task, we must process the above data. By linear interpolation or age structure, the proper data produce. Concretely speaking, when the interval in some group is narrow, the linear interpolation is employed; in the contrast, especially when the age structure is very similar, we will turn to the analogy method. At the same time, gender ratio for each group is important for us. Similarly, we could estimate it by seek for rule. By using above methods, we have the result:

Table 3

The data by processing

Age	0-4	5-9	10-14	15-19	20-24	25-29	30-34
Total	388	51	64	83	1284	4714	7151
Male	202	35	49	69	1116	4181	6458
Female	186	16	15	14	168	533	693
Age	35-39	40-44	45-49	50-54	55-59	Over 60	
Total	6232	3880	2198	1273	817	859	
Male	5778	3663	2100	1207	767	714	
Female	454	217	98	66	50	145	

At the same time, we find many interesting conclusions. We will illuminate them one by one. Firstly, the age structure of newly diagnosed cases every year varies slowly as the time goes on. The similar figure is below:

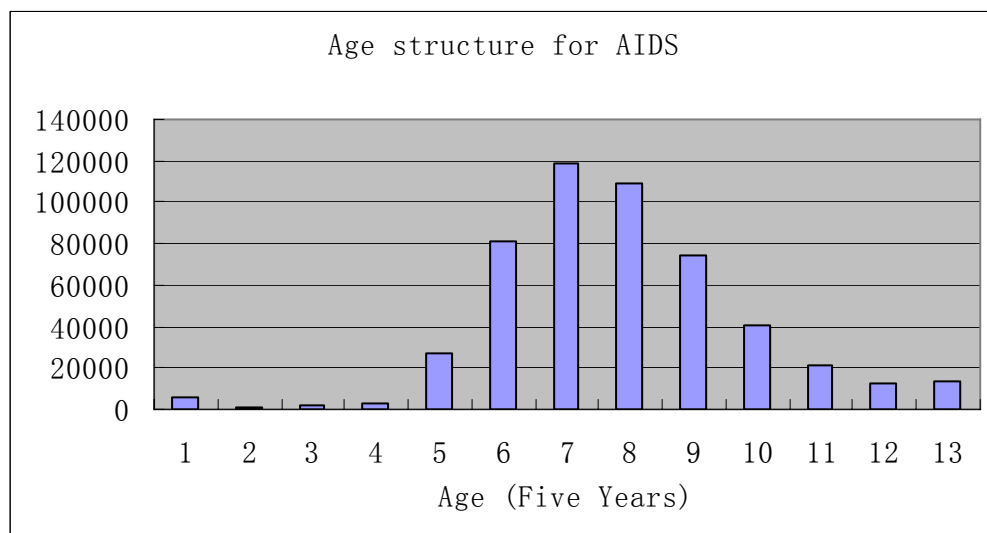
**Figure 2.** Every group ratio in the amount

Figure 2 shows that the amount with age from 0 to 20 is few. If we consider the method of HIV/AIDS transmission, it will be obviously fact. The amount of age 0-4 is more than other age 5-9, or 10-14, and 15-19. This reason is that the infant will acquire HIV' source from his or her mother, which is very important aspect.

Then the gender ratio changes as the age group varies, but changes slowly as the time goes on. The figure is below:

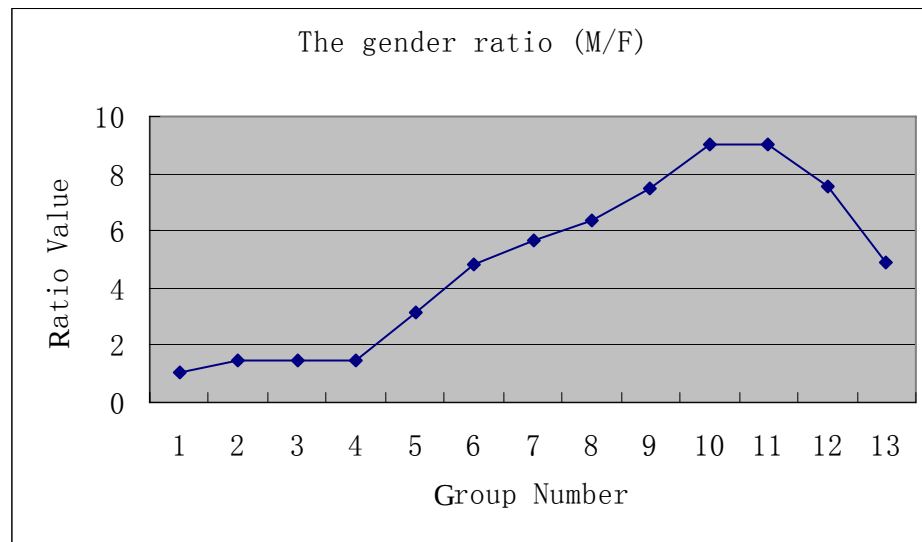


Figure 3. The gender ratio for USA AIDS diagnosed in year 1992.

This figure is one of many years, and they have common feather. From age 0 to 20, male and female are similar about AIDS. However, as age increases continually, the difference comes out. The main reason is that male have free, enthusiasm, and pizzazz, female have conservative, virtuous and so on. So the gender ratio is far large during the most times.

In fact, our purpose is to estimate the infection rate for each group. So we introduce the definition of infection rate ρ firstly:

$$\rho = \frac{\Delta p}{hI}$$

Where ρ is infection rate during some time, Δp is increment on the amount of AIDS during that time, h is the amount of corresponded healthy people, and I is the amount of all AIDS at the begin of that time.

We utilize the definition, by combining it with the above data that was processed, to determine many ρ for their own group. When the interval is five years, it is properly desired. In fact, *infection rate* varies as the time goes on; but now data points are limited, we have to choose the average method. In the

predicting the number of AIDS process, we will adjust it to adapt more conditions. By making program in MATLAB, we find that they vary with gender and age. The following figure will show you this feather:

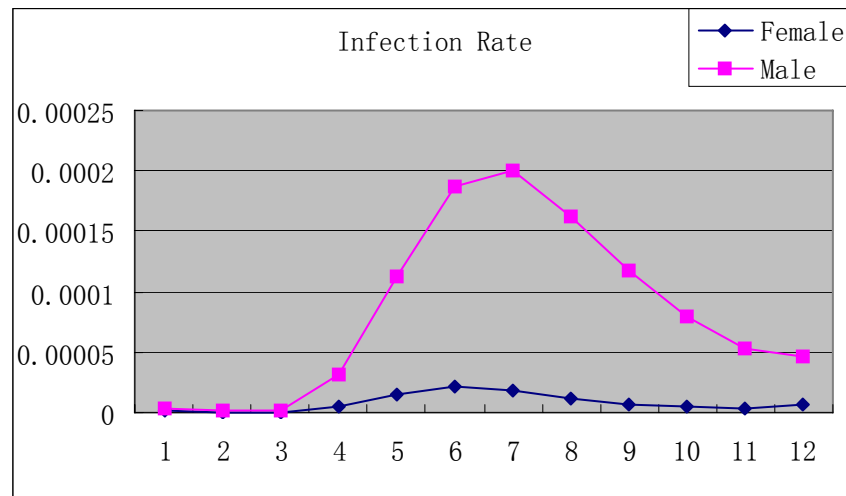


Figure 4. Each group's infection rate

Figure 4 shows that Male's *infection rate* is far greater than Female's from age 20 to 50. It is clear that the result comes out. However, Infection rate-estimating is a hard work. The determination of certain parameters for the other 5 countries is the same as the determination of America.

Financial resources estimating model

In order to estimate the financial resources to against HIV/AIDS, we first consider how to control the spread of HIV/AIDS.

Control factors

There are a number of interventions that HIV/AIDS could be directed towards – including preventions interventions and care interventions.

Preventions interventions: Aim to protect people living with AIV/AIDS; we take preventions interventions such as voluntary counseling, condom social marketing, school-based AIDS and so on. In this way, a large of healthy people can avoid infecting.

Care interventions: In order to take care of AIV/AIDS infections, we take measures to prolong these persons' life such as treating other untreated sexually transmitted diseases, treating opportunistic and so on. In this way, we can reduce AIDS/HIV source of infection.

Different interventions will lead to different impacts to AIDS control. We consider the relationship between preventions interventions and care interventions to direct our interventions.

Control model

In previous model, we forecast the AIDS/HIV infections in the absence of any additional

interventions. Now we build a model to the rate of change in the number of HIV/AIDS infections for USA with preventions interventions and care interventions.

Supposition:

The population in USA is a constant in a certain time period. Pay no attention to birth rate, mortality rate, immigration or emigration.

We only consider healthy people and AIV/AIDS infections in the USA.

$i(t)$: The percentage of HIV/AIDS infections in USA population at t time.

$h(t)$: The percentage of healthy people in USA population at t time.

We can get a conclusion here:

$$i(t) + h(t) = 1$$

λ : healthy persons who will become infections per unit time per infection.

N : the total population of USA

According to the supposition before and in the absence of interventions, per HIV/AIDS patient will infect $\lambda h(t) N$ healthy persons per unit time. So $\lambda h(t) N i(t)$ healthy persons will be infected per unit time. The number of patient is Ni , and $\lambda N h i$ represents its increasing rate. So

$$N \frac{di}{dt} = \lambda N h i \quad .(1)$$

and

$$i(t) + h(t) = 1$$

We will take measures to control HIV/AIDS by preventions interventions and care interventions.

Prevention interventions protect healthy persons. We suppose that $k_2 h N$ can avoid infecting. k_2 represent the level of Prevention interventions

Care interventions reduce the number of infection source. We suppose that the number of infection source reduce to $k_1 i$. k_1 represent the level of Prevention interventions

So (1) change to :

$$N \frac{di}{dt} = \lambda N h (1 - k_1) i - k_2 N h \quad (2)$$

We determine whether i reach equilibrium level. The rest point or equilibrium levels occur when

$$\frac{di}{dt} = 0$$

Setting the right side of equations (2) equal to zero and solving for i give the rest point $i = 1$ and

When $i = 1$, the whole country become a AIDS country, no one is healthy.

When $i = \frac{k_2}{\lambda(1-k_1)}$, the growth $\frac{di}{dt}$ in the number of HIV/AIDS is zero.

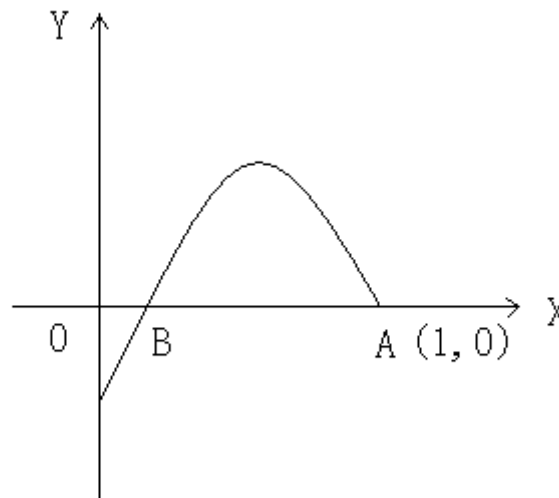


Figure 5

From the figure, we can see that there are two rest points. The right rest point is $A(1,0)$, where there is no healthy person, i represent the percentage of HIV/AIDS infections in USA population, all country becomes a AIDS country. We are sad to see this result. The left rest point is $B(\frac{k_2}{\lambda(1-k_1)}, 0)$, if we want to control the growth of HIV/AIDS, we need to reduce i to the left of B.

The rest point B depends on three parameters λ, k_1, k_2 . We can strengthen our control on the spread of AIDS by controlling the two factors: k_1 and k_2 . The more to the right that Point B displayed, the stronger we control the spread of AIDS.

k_2 and k_1 represent the level of Prevention interventions and care interventions. Both k_1 and k_2 less than 1. The more we increase k_1 and k_2 , the less the number of HIV and AIDS will increase.

Conclusion:

We should pay much attention to a qualitative graphical analysis for solution to differential equations. We can find some useful conclusions to control spread of HIV/AIDS.

Only considering prevention interventions

We take measures to protect healthy persons only, and ignore HIV/AIDS infections. In other words,

k_1 will change to zero in our model.

$$\frac{di}{dt} = \lambda hi - k_2 h \quad (k_1 = 0)$$

We can reduce the number of infection by the right prevention interventions level.

If

$$k_2 h > \lambda hi$$

Then

$$\frac{di}{dt} < 0 \quad (3)$$

The expression shows that the number of HIV/AIDS would decrease.

In order to control the spread of HIV/AIDS, we should try our best to develop vaccine.

Only considering care interventions

We pay much attention to look after infection patients. We hope to control source of infection to reduce the growth of infections.

$$\frac{di}{dt} = \lambda(1 - k_1)hi \quad (k_2 = 0)$$

We are able to reduce the growth of infections without the limitation of prevention interventions. But we can't reduce the number of infection patients.

Both prevention interventions and care interventions

From the analysis above, both prevention interventions and care interventions is also a useful method to control the HIV/AIDS infection.

These conclusions can direct our treatment to control HIV/AIDS.

Application of the Conclusion

HIV Treatment in Poor Countries

Limited by financial resource, poor countries should spend much money in improving prevention interventions, which can reduce the growth of infections effectively. So when we consider the financial resource to poor countries, we assume that all money is paid to prevention interventions.

HIV Treatment in High-Income Countries

High-income countries have proven the success to against HIV/AIDS. They fight against HIV/AIDS by both prevention interventions and care interventions. In wealthy countries, there has been dramatic success in the fight against HIV/AIDS, success that has been largely achieved through the use of antiretroviral therapy.

As the problem state told us, we only focus on two potential interventions: ARV drug therapy belong to care interventions while vaccine belong to prevention interventions. But as effective vaccine has not been developed, we research it's affection solely.

Vaccine

A preventive vaccine is the best hope for ending the AIDS epidemic. But unless the vaccine is widely available and used by those at risk of HIV, this goal will not be realized. So we make some assumptions about HIV Vaccine according the quality of the diphtheria-pertussis-tetanus vaccine

Assumption:

1. HIV/AIDS preventative vaccine is developed in 2017.
2. The steady-state level for new cohorts immunization rates equals to the third dose of the diphtheria-pertussis-tetanus vaccine in every country.
3. The steady-state level for adults (any group over age 5) immunization rates equals to the second dose of the diphtheria-pertussis-tetanus vaccine in every country.
4. The vaccine can protect 95% users and last efficacy for 5 years.
5. There is no epidemiological externalities from vaccination
6. The price of the vaccine is \$3 per dose.

Financial resources estimating model

We propose a model to evaluate the financial resources to six countries (China, English, Nigeria, Mexico, Australia, USA).

Because of limited time and complex of the problem, we make assumption as follow:

1. The financial resources to against HIV all from foreign aid donors by each country.

The six countries lay in different GNP levels classified by World Bank. Different countries will get different financial resources levels. It is too complex to confirm the percentage. And some countries such as USA, English, Australia may not need financial resources. So we make such assumption.

2. Each country does not change treatment to HIV/AIDS.

For example, Mexico will provide ARV to 92% of AIDS/HIV infections in 2002. As time on, Mexico do not change the percentage in any other year. We deal with the steady-state level for HIV vaccine in the same way.

3. Each country will get financial resources they need.

$i = 1, 2, 3, \dots, 6$: 1, 2, 3, ..., 6 refer to China, English, Nigeria, Mexico, Australia, USA.

t : represents to the year.

$S_i(t)$: S represents to the total cost to against HIV/AIDS.

$A_i(t)$: A represents to the total cost of ARV .

$V_i(t)$: V represents to the total cost of vaccine

The financial resources spend to against HIV is divided into two parts, the cost for vaccine and the cost for ARV each year.

$$S_i(t) = A_i(t) + V_i(t)$$

We change the parameters of model for Task 1. According to the feather of our model, we firstly estimate the amount of ADIS cases, and then determine the total cost for each selected country. Part of results is below:

Table 4
Financial demand for selected countries

Country	USA	China	Nigeria	Mexico	Australia	UK
2007	1.325	5.168	50.59	0.747	0.912	0.409
2012	1.654	9.928	250	0.676	2.678	0.465
2017	2.013	20.108	851	0.579	5.65	0.472
2022	2.315	32.475	1490	0.512	10.01	0.464

Improved Model considering ARV-resistant disease strains

There are disease strains during the ARV drug treatment. We attribute its impact to some parameters such as death rate, the cost to ARV treatment. Some patient may lose their lives by disease strain, the death rate of AIDS will change. We can adjust this parameter by comparing another sort of disease. Because of the disease strains, another patient may need second-line or third-line ARV drugs. The cost to treatment will change. We adjust cost by the price of second-line and third-line cost.

Our model is flexible, we can adjust parameters according to the impact from disease strains. It is very pity to fail to give the result, because of the limitation of time.

A white paper providing recommendations

The worldwide AIDS pandemic continues to gather force. By the end of the year, an estimated 40.3 million people worldwide are infected with HIV and face disease and early death unless they receive appropriate life-extending medical care. More than 25 million people have died of AIDS since 1981.

In addition to tremendous human suffering, the pandemic has become a major cause of social, political and economic instability.

To control the spread of HIV/AIDS is a war against an enemy that is devastating, its peoples, economies, and societies. And it goes beyond. It is truly a global issue.

As the resources available are limited, how to allocate them to achieve a best effect becomes a big problem. Therefore, we create a model to solve this problem.

Here are some recommendations:

- 1) Poor countries should spend much money in improving prevention interventions, which can reduce the growth of infections effectively. So when we consider the financial resource to poor countries
- 2) High-income countries can fight against HIV/AIDS by both prevention interventions and care

interventions. In wealthy countries.

As an international concern, working to end AIDS is a public health imperative that requires a comprehensive set of strategies to address the needs of people living with and at risk for HIV. Relative to other foreign policy priorities, combine as many as countries to control the spread of HIV/AIDS is a very important foreign policy for many countries.

HIV/AIDS and associated diseases kill as many as a quarter of the population in the hardest hit countries, and the emergence and re-emergence of more virulent and antimicrobial resistant strains, pose clear threats to some of the most vulnerable nations and regions. The devastating impact of diseases like HIV/AIDS is crushing political, economic, and social development in many countries; development that is crucial to our efforts to promote increased security and stable governments, trade, democratic development, and a number of other key foreign policy priorities.

Simply put, health issues threaten the very development and cornerstones that are necessary for building and maintaining an international community that is stable, democratic, and economically prosperous. So promoting international health is one of the Department of State's strategic goals.

The scope of HIV/AIDS is truly unprecedented, and it threatens to reverse decades of political development, economic growth, and social progress. We cannot underestimate the devastating impact of the HIV/AIDS epidemic on refugees -- especially in Africa.

Even though Africa is truly at the center of this pandemic, we should focusing our attention and resources globally to include Asia, Russia, and the New Independent States, and parts of the Western Hemisphere -- especially the Caribbean. Diseases obviously know no boundaries, so we should target our efforts globally.

Because this is not a battle that we can or should fight alone, every country should working with each other. It is imperative that each member matches financial commitment to fighting HIV/AIDS.

The State Department should Government's foreign policy to combat the spread of HIV/AIDS. Each country should joining to the interagency development of a strategy to expand HIV/AIDS engagement beyond Africa to other high-threat regions, including Southeast Asia and China, India, Russia, Central Asia, and the Caribbean. It's also necessary to ensure that funding is programmed effectively in terms of national interest, humanitarian need, impact on social and economic sectors, and other criteria. There is a taskforce that deals with this issue.

Every country must contribute to this fund for the fund to be effective. It cannot work without full support from other members of the international community.

All of this effort and financial support is an indication of the Administration's serious commitment to fighting HIV/AIDS and a recognition of the impact they have on our national security interests and other foreign policy priorities.

To coordinate donor so as to control the spread of HIV/AIDS on world wide, expect considering the AIDS spreading in one country, the economic position should also be considered.

For developed countries, we do not allocate any donor for them because they can apply enough financing by themselves.

For developing countries, we prorate donor according to proportion of the financial demand.

Strengths

- **The age structure and gender ratio** about HIV/AIDS cases are involved in our model. It will

make our prediction and analysis about future of HIV/AIDS cases more accurately.

- **The transmission methods** of HIV/AIDS are considered in our model, such as mother-infant HIV transmission, and others including Injecting drug use, sex worker.
- **Differential equation** is employed to discuss deeply the fact of different intervention strategy and propose some useful advice.

Weakness

- The estimation and adjustment about infection rate are not very accurate.
- The allocation of financial resources is not deeply discussed.

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Appendix

Data information extracted from Original data about AIDS for USA from 1981 to

1992

Original data come from the following Web.

<http://www.cdc.gov/hiv/topics/surveillance/resources/reports/past.htm#surveillance>

Year 1981

Age	Total	Male	Female
Under5	5	3	2
5-9	0	0	0
10--14	0	0	0
15--19	1	1	0
20-24	15	13	2
25-29	55	49	6
30-34	84	76	8
35-39	73	68	5
40-44	45	42	3
45-49	26	25	1
50-54	15	14	1
55-59	10	9	1
Over60	10	8	2

Year 1982

Age	Total	Male	Female
Under5	6	3	3
5-9	0	0	0
10--14	1	1	0
15--19	1	1	0
20-24	20	17	3
25-29	73	65	8
30-34	111	100	11
35-39	97	90	7
40-44	60	57	3
45-49	34	32	2
50-54	20	19	1
55-59	13	12	1
Over60	13	11	2

Year 1983

Age	Total	Male	Female
-----	-------	------	--------

Under5	41	22	19
5-9	5	4	1
10--14	7	6	1
15--19	9	8	1
20-24	135	118	17
25-29	498	442	56
30-34	756	683	73
35-39	658	610	48
40-44	410	387	23
45-49	232	222	10
50-54	135	128	7
55-59	86	81	5
Over60	91	76	15

Year 1984

Age	Total	Male	Female
Under5	102	54	48
5-9	13	9	4
10--14	16	12	4
15--19	21	18	3
20-24	337	293	44
25-29	1236	1097	139
30-34	1876	1695	181
35-39	1635	1516	119
40-44	1018	961	57
45-49	576	551	25
50-54	334	317	17
55-59	214	201	13
Over60	225	187	38

Year 1985

Age	Total	Male	Female
Under5	213	111	102
5-9	28	19	9
10--14	35	27	8
15--19	46	38	8
20-24	706	614	92
25-29	2592	2299	293
30-34	3932	3551	381
35-39	3427	3178	249

40-44	2133	2014	119
45-49	1208	1155	53
50-54	700	664	36
55-59	449	422	27
Over60	472	393	79

Year 1986			
Age	Total	Male	Female
Under5	388	202	186
5-9	51	35	16
10--14	64	49	15
15--19	83	69	14
20-24	1284	1116	168
25-29	4714	4181	533
30-34	7151	6458	693
35-39	6232	5778	454
40-44	3880	3663	217
45-49	2198	2100	98
50-54	1273	1207	66
55-59	817	767	50
Over60	859	714	145

Year 1987			
Age	Total	Male	Female
Under5	667	347	320
5-9	88	59	29
10--14	110	83	27
15--19	143	118	25
20-24	2203	1913	290
25-29	8086	7172	914
30-34	12265	11076	1189
35-39	10689	9910	779
40-44	6655	6283	372
45-49	3770	3602	168
50-54	2184	2071	113
55-59	1402	1316	86
Over60	1474	1224	250

Year 1988

Age	Total	Male	Female
Under5	1110	579	531
5-9	147	99	48
10--14	184	138	45
15--19	239	197	42
20-24	3666	3184	482
25-29	13454	11933	1521
30-34	20408	18430	1978
35-39	17786	16490	1296
40-44	11074	10454	620
45-49	6274	5994	280
50-54	3635	3446	189
55-59	2333	2190	143
Over60	2453	2038	415

Year 1989

Age	Total	Male	Female
Under5	1643	850	793
5-9	220	145	75
10--14	263	191	71
15--19	329	262	67
20-24	5090	4346	744
25-29	18966	16744	2222
30-34	28871	25927	2944
35-39	25463	23422	2041
40-44	15960	14957	1003
45-49	8991	8531	460
50-54	5162	4874	288
55-59	3318	3085	233
Over60	3504	2922	582

Year 1990

Age	Total	Male	Female
Under5	2276	1169	1107
5-9	318	207	111
10--14	370	259	111
15--19	449	337	112

20-24	6720	5652	1068
25-29	25634	22531	3103
30-34	38895	34748	4147
35-39	35132	32034	3098
40-44	22371	20799	1572
45-49	12618	11892	726
50-54	7021	6561	460
55-59	4461	4107	354
Over60	4806	3998	808

Year 1991

Age	Total	Male	Female
Under5	2802	1449	1353
5-9	418	265	153
10--14	476	323	152
15--19	563	412	151
20-24	8160	6765	1395
25-29	32202	28080	4122
30-34	49316	43715	5601
35-39	45404	41064	4340
40-44	29454	27202	2252
45-49	16547	15464	1083
50-54	9141	8456	685
55-59	5691	5196	495
Over60	6217	5177	1040

Year 1992

Age	Total	Male	Female
Under5	3432	1754	1678
5-9	510	314	196
10--14	576	380	196
15--19	675	479	196
20-24	9582	7820	1762
25-29	38713	33503	5210
30-34	59865	52809	7056
35-39	56053	50309	5744
40-44	37216	34047	3169
45-49	20771	19303	1468
50-54	11450	10547	903
55-59	6891	6251	640

Over60	7712	6454	1258
--------	------	------	------

The program used for estimating parameters death rate, infection rate

```
% Data.mat used for provide information
%Estimate some parameters such as death rate, infection
%rate five years group.
%Used data about 1981-1992 from USA

% Unit: thousands
clc,clear;
format long;
load Data
%Estimate infection rate
InfectionRate=zeros(24,8);
%NewlyYear=zeros(24,13); %M,F;from 1981 to 1992.
State_before1981=0.07;
%A=zeros(4,12);
%B=zeros(4,12);
% find the data by interplore from 1981 to 2002.
AgeMale=zeros(12,12);
AgeFemale=zeros(12,12);
for i=1:12
if rem(i,5)
AgeMale(i,:)= A(floor(i/5)+1,:)+rem(i,5)*(A(floor(i/5)+2,:)-A(floor
(i/5)+1,:))/5;
AgeFemale(i,:)=B(floor(i/5)+1,:)+rem(i,5)*(B(floor(i/5)+2,:)-B(floor
(i/5)+1,:))/5;
Else
AgeMale(i,:)= A(i/5,:);
AgeFemale(i,:)=B(i/5,:);
end
end
HivState=zeros(1,13);
HivState(1)=State_before1981;
% Compute initial state every year.
for i=2:13
HivState(i)=HivState(i-1)+UsaNewly(i-1)-UsaDeath(i-1)
end
% Compute Infection Rate.
for i=1:24
for j=1:8
if rem(i,2)
```

```

        % Male
InfectionRate(i,j)=(0.8*NewlyYear(2*j-1,(i+1)/2)+0.2*NewlyYear(2*j-1,
(i+1)/2+1)+0.6*NewlyYear(2*j+1,(i+1)/2)+0.4*NewlyYear(2*j+1,
(i+1)/2+1)
+0.4*NewlyYear(2*j+3,(i+1)/2)+0.6*NewlyYear(2*j+3,(i+1)/2+1)+0.2*
NewlyYear(2*j+5,(i+1)/2)+0.8*NewlyYear(2*j+5,(i+1)/2+1)+NewlyYear
(2*j+7
,(i+1)/2+1))/(HivState(j+3)*AgeMale(j+2,(i+1)/2));
        else
            % Female
InfectionRate(i,j)=(0.8*NewlyYear(2*j,i/2)+0.2*NewlyYear(2*j,i/2+1)
+0.6
*NewlyYear(2*j+2,i/2)+0.4*NewlyYear(2*j+2,i/2+1)+0.4*NewlyYear
(2*j+4,i/2)+
0.6*NewlyYear(2*j+4,i/2+1)                                +0.2*NewlyYear(2*j+6,i/2)
+0.8*
NewlyYear(2*j+6,i/2+1)                                +NewlyYear(2*j+8,i/2+1))/
(HivState(j+3)*AgeFemale(j+2,i/2));
        end
    end
end

%Estimate death rate
%UsaNewly=zeros(1,12);
%UsaDeath=zeros(1,12);
DeathRate=zeros(1,8);
for i=1:8
    DeathRate(i)=sum(UsaDeath(i:i+4))/(HivState(i)+sum(UsaNewly
(i:i+4)));
end
DeathRate=mean(DeathRate);
DeathRate1=zeros(1,12);
for i=1:12
    DeathRate1(i)=UsaDeath(i)/(HivState(i)+UsaNewly(i));
end
for i=1:24
    if rem(i,2)
        InfectionMale((i+1)/2,:)=InfectionRate(i,:);
    else
        InfectionFemale(i/2,:)=InfectionRate(i,:);
    end
end
end

```

The program used for predict the information about infection-HIV/AIDS from 2006 to 2050.

```
% Data.mat used for provide information
% Predicting the information about infection-HIV/AIDS from 2006 to
2050 For USA

% Unit: thousands
clc,clear;
format long;
load Data
%-----
PredictData=zeros(13,2,10); % Apply 3D-Matrix.
FactorMean=0.05; % To be determined.
DeathRate=0.40;
Population=zeros(1,10);
Population(1)=289820.909; % 2002
Fertility=Fertility/1000;
% N=5000;
PopulationPre=[298212.895,312253.262,325723.414,338426.99,350102.576,
360893.731,370708.985,...
379543.693,387530.913,394976.485];
for i=1:9
    PP(i)=(PopulationPre(i)+PopulationPre(i+1))/2;
end
InfectionFemale=[2.08378E-06,6.89071E-07,6.51286E-07,5.32546E-06,
0.000015609, 2.18081E-05,
1.82808E-05,1.12906E-05,6.37116E-06,4.4142E-06,
3.46884E-06,7.41966E-06];
InfectionMale=[2.64481E-06,1.77E-06,2.42E-06,3.08E-05,
1.13E-04,1.87E-04, 2.00E-04,1.62E-04,
1.17E-04,7.97E-05,5.30E-05,4.62E-05];
%Vary to simulate
InitialStateMale=[2.058,0.393,0.757,2.807,23.068,69.208,99.839,
94.473,64.408,34.014,15.928,7.429,4.464];
InitialStateFemale=[1.987,0.247,0.472,1.746,6.684,13.07,16.382,
13.471,7.557,3.302,1.594,0.903,0.908];
PredictData(:,1,1)=InitialStateMale;
PredictData(:,2,1)=InitialStateFemale;
% Let's go!
for i=1:9
    Factor=FactorMean+randn*0.005/3;
```

```

%InfectionFemale=InfectionFemale*(1-sum(sum(PredictData(:, :, i)))/N);
%InfectionMale=InfectionMale*(1-sum(sum(PredictData(:, :, i)))/N);
    InfectionFemale=InfectionFemale*(0.83);
    InfectionMale=InfectionMale*(0.83);
    for j=1:12
        InMale(j)=max(InfectionMale(j)*(Population(i)/(PP(i))
            *AgeMalePre
            (i,j)-PredictData(j,1,i))*
            sum(sum(PredictData(:, :, i)),0);
        InFemale(j)=max(InfectionFemale(j)*((Population(i)/PP(i))
            *AgeFemalePre
            (i,j)-PredictData(j,2,i))*sum(sum(PredictData(:, :, i)),0);
        end TempMale=(1-DeathRate)*(PredictData(1:12,1,i)+InMale');
        TempFemale=(1-DeathRate)*(PredictData(1:12,2,i)+InFemale');
        InfantMale=0.5*0.8*0.25*sum(PredictData(4:10,2,i).*Fertility(i,:));
        InfantFemale=InfantMale;
        PredictData(1,1,i+1)=InfantMale;
        PredictData(1,2,i+1)=InfantFemale;
        PredictData(2:13,1,i+1)=TempMale;
        PredictData(2:13,2,i+1)=TempFemale;
        DeathNum=sum((PredictData(1:12,1,i)+InMale'+PredictData(1:12,2,i)
            +InFemale'))
        *DeathRate);
        Population(i+1)=Population(i)*(1+Factor)-DeathNum;
        All(i)=sum(sum(PredictData(:, :, i)));
    End

```

The program used for predict the future about AIDS considering ARV, Vaccine and drug-resistant strains of HIV

```

% DataNew.mat used for provide information
% Predicting the information about infection-HIV/AIDS from 2006 to
2050 For USA
% Unit: thousands
clc,clear;
format long;
load Data
%-----
%AgeMalePre=zeros(10,12);
%AgeFemalePre=zeros(10,12); % 2005-2050, 0-60.
%Fertility=zeros(9,7); % 2005-2050,15-49.
PredictData=zeros(13,2,10); % Apply 3D-Matrix.
FactorMean=0.05; % To be determined.

```

```

DeathRate=(0.532*0.2+0.468*0.4);
Population=zeros(1,10);
Population(1)=289820.909; % 2002
Fertility=Fertility/1000;
% N=5000;
PopulationPre=[298212.895,312253.262,325723.414,338426.99,350102.576,
360893.731,37070.985,...
379543.693,387530.913,394976.485];
for i=1:9
PP(i)=(PopulationPre(i)+PopulationPre(i+1))/2;
end
InfectionFemale=[2.08378E-06,6.89071E-07,6.51286E-07,5.32546E-06,
0.000015609,2.18081E-05,1.08E-05,1.12906E-05,6.37116E-06,
4.4142E-06,3.46884E-06,7.41966E-06];
InfectionMale=[2.64481E-06,1.77E-06,2.42E-06,3.08E-05,1.13E-04,1.87E-
04, 2.00E-04,1.62E-04,1.17E-04,7.97E-05,5.30E-05,4.62E-05];
InitialStateMale=
[2.058,0.393,0.757,2.807,23.068,69.208,99.839,94.473,
64.408,34.014,15.928,7.429,4.464];
InitialStateFemale=
[1.987,0.247,0.472,1.746,6.684,13.077,16.382,13.471,
7.557,3.302,1.594,0.903,0.908];
PredictData(:,1,1)=InitialStateMale;
PredictData(:,2,1)=InitialStateFemale;
% Let's go!
for i=1:9
    Factor=FactorMean+randn*0.005/3;
%InfectionFemale=InfectionFemale*(1-sum(sum(PredictData(:, :, i)))/N);
%InfectionMale=InfectionMale*(1-sum(sum(PredictData(:, :, i)))/N);
InfectionFemale=InfectionFemale*(0.83);
InfectionMale=InfectionMale*(0.83);
    if i>=3
        for j=1:12
            if j==1
InMale(j)=max(0.468*(0.073+0.927*0.05)*InfectionMale(j)*
(Population(i)/(PP(i))*AgeMalePre(i,j)-PredictData(j,1,I ))*sum
(sum(PredictData(:, :, i))),0);
InFemale(j)=max(0.468*(0.073+0.927*0.05)*InfectionFemale(j)*
((Population
(i)/PP(i))*AgeFemalePre(i,j)-PredictData(j,2,i))*sum(sum(PredictData
(:, :, i))),0);
            else
                InMale(j)=max(0.468*(0.34*0.05+0.66)
*InfectionMale(j)*(Population(i)/(PP(i))*

```

```

AgeMalePre(i,j)-PredictData(j,1,i))*sum(sum(PredictData(:, :, i))), 0);
    InFemale(j)=max(0.468*(0.34*0.05+0.66)*InfectionFemale(j)*
    ((Population(i)
/PP(i))*AgeFemalePre(i,j)-PredictData(j,2,i))*sum(sum(PredictData
(:, :, i))), 0);
    end
    end
    else
    for j=1:12
    InMale(j)=max(0.468*InfectionMale(j)*(Population(i)/(PP(i))*
AgeMalePre(i,j)-PredictData(j,1,i))*sum(sum(PredictData
(:, :, i))), 0);
    InFemale(j)=max(0.468*InfectionFemale(j)*((Population(i)/PP(i))
*AgeFemalePre(i,j)-PredictData(j,2,i))*sum(sum(PredictData
(:, :, i))), 0);
    end
    end
    TempMale=(1-DeathRate)*(PredictData(1:12,1,i)+InMale');
    TempFemale=(1-DeathRate)*(PredictData(1:12,2,i)+InFemale');
    InfantMale=0.5*0.8*0.25*sum(PredictData(4:10,2,i).*Fertility(i,:));
    InfantFemale=InfantMale;
    PredictData(1,1,i+1)=InfantMale;
    PredictData(1,2,i+1)=InfantFemale;
    PredictData(2:13,1,i+1)=TempMale;
    PredictData(2:13,2,i+1)=TempFemale;
    DeathNum=sum((PredictData(1:12,1,i)+InMale'+PredictData(1:12,2,i)+
    InFemale')*DeathRate);
    Population(i+1)=Population(i)*(1+Factor)-DeathNum;
    All(i)=sum(sum(PredictData(:, :, i)));
    end

```